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# A Dynamic Spike Sorter for Multiscale Nanoelectrode Array Recordings

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**High-throughput, intracellular electrophysiology is crucial for advancing the understanding of neuronal processing and network dynamics. Nanoelectrode arrays (NEA) offer a promising approach by directly capturing intracellular signals across sub-neuronal compartments, including action potentials, postsynaptic potentials, and low-frequency membrane fluctuations. However, the complexity of NEA datasets, characterized by multiscale events of varying amplitude and duration, demands novel analytical strategies. In this work, a dynamic spike sorting pipeline is introduced and designed to isolate, extract, and sort these diverse electrical signals within a landscape of spontaneous electrical behavior. It is obtained estimates of signal attenuation and distortion using a bespoke biophysical circuit simulation designed to match the specific nanoelectrode interface. Based on these observations, bounds are set for filtering and extracting multiscale waveforms, and validated their isolation using pharmacological data. Finally, it is shown that multiscale analysis of spontaneous electrical recordings reveals interrelationships between high frequency events such as action potentials, and low frequency membrane potential fluctuations which may inform models of neuronal network excitability. Advanced sorting algorithms tailored for nanoelectrode array recordings are essential for unlocking the full potential of next generation, high throughput neuroelectronic devices and achieving a deeper understanding of neuronal dynamics.**

## 1. Introduction

### 1.1. Multiscale Recording from NEAs

Mapping neuronal activity at multiple scales is essential for unraveling how the brain processes information, from subcellular events at synapses to large-scale network dynamics. Intracellular recordings capture both suprathreshold action potentials and subthreshold fluctuations, offering a comprehensive view of synaptic activity and ionic processes that shape neural function[1–4] and changing ionic roles throughout development.[5–11] Recording spontaneous native activity from multiple neurons simultaneously can reveal the orchestration of these events, insights that traditional evoked protocols cannot provide.[12–18] Nanoelectrodes arrays (NEAs) are emerging as powerful tools for this purpose, offering high-throughput recordings with exceptional spatial and temporal resolution.[19,20] NEAs employ 3D nanostructured electrodes that couple tightly with single cells,[21–24] allowing them to record intracellular waveforms from single neurons that were previously detectable only by low-throughput patch clamp techniques. However, NEAs can also be massively parallelized to record from thousands of adherent cells simultaneously, toward applications in drug discovery or neuronal network modeling.

Nevertheless, the vast amount of information encoded in NEA recordings presents a new challenge: interpreting complex data structures, which differ significantly from traditional electrophysiological recordings. Currently, no robust algorithms exist to fully interpret these signals. Existing standard spike sorters

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present significant barriers to analyzing the intracellular, multiscale NEA signals. These datasets contain diverse signals, including action potentials (APs), postsynaptic potentials (PSPs), and low frequency membrane fluctuations (LFFs). The AP, PSP, and LFF waveform shapes exhibit varying amplitudes (hundreds of microvolts to tens of millivolts) and durations (tens to hundreds of milliseconds). Such diverse events are not compatible with standard spike sorting algorithms designed for low amplitude, few-millisecond long APs from putative single "units" recorded using an extracellular electrode (>10 µm diameter).[25,26] The extracellular AP consistency allows conventional spike sorters to uniquely assign spikes to single units (which putatively represent single neurons) based on their spike waveform through pattern matching techniques such as K-means and supra-paramagnetic clustering.[25,27–29] However, NEA datasets cannot rely on similar waveform shapes, since they possess multiscale signals which are dynamic throughout the recording.

Thus far, several groups have presented significant progress toward dynamic spike sorters for NEA datasets. Seminal studies performed by early researchers in the field have provided insight into the identity of intracellular signals that can be measured using nanoelectrodes of various sizes, shapes, and materials.[22,24,30–34] Manual identification of supra- and sub-threshold signals from nano-functionalized MEAs using ground truth patch-clamp electrophysiology enabled spike sorting across sub-neuronal compartments.[35] Spike-triggered averaging across thousands of nanoelectrodes allowed for precise identification of AP, excitatory PSP (EPSP), inhibitory PSP, and AP + EPSP templates to enable functional connectivity mapping for a significant proportion of in vitro neuronal networks.[20] Furthermore, several groups including ours have demonstrated the "quantal" nature of PSPs which can be used to estimate synaptic strength in real time.[1,18,35] It was recently shown that applying conventional spike sorting to intracellular NEA signals can provide insights into standard network metrics such as coherence and cross correlation.[36] However, this approach treats multiscale signals equally, causing information loss since the relationships between subthreshold synaptic activity cannot be delineated from supra-threshold neuron-neuron communication. The main drawback of existing NEA analysis tools has been that experiments used to obtain ground truth data to validate multiscale signals (APs, PSPs) – specifically, current injection into single neurons – inherently limits the spectrum of multiscale signals obtained. Here, using datasets from spontaneous NEA recordings of 2D and 3D neuronal networks validated using pharmacology,[18] we propose a pipeline to analyze 3 different classes of multiscale signals and create topological maps of neuronal network activity.

Our main objectives were to 1) isolate multiscale signals such as APs, PSPs, and LFFs without distorting their waveform structure as an alternative means of event detection and signal processing, 2) present a circuit model to describe expected trends in electrical coupling, signal attenuation, and distortion which have not previously been provided for our nanoelectrodes, and 3) understand how pharmacology and non-spike triggered recording experiments could be analyzed to draw biological conclusions about multiscale network behavior. We showed that the isolation, extraction, and analysis of multiscale signals informed by biophysical models could be used to create 3D topological maps of network activity within 2D and 3D neuronal cultures. Specifically, we showed for the first time that spontaneous (non spike-triggered) NEA recordings can achieve a broader spectrum of multiscale signals, providing an information rich bioelectrical landscape to describe neuronal networks in both 2D and 3D. Developing new analytical approaches to decode multiscale, intracellular recordings could provide unprecedented insights into network activity and brain function. Obtaining such datasets from multiple dendritic compartments from many neurons at once could paint a picture of network activity with subcellular resolution. Particularly, neuroscientists could obtain topological maps of subthreshold and network activity, directly tying synaptic and metabolic events to network changes[37–39] and also clarifying how synaptic inputs are processed by single neurons.[40–42]

### 1.2. Conventional Spike Sorters Fall Short of Multiscale Spike Sorting

Single neurons are commonly depicted as having two types of transient membrane potential events: APs, a self-propagating pulse that travels along the axon and triggers synaptic transmission at the synapse, and PSPs, low voltage membrane potential events associated with synaptic transmission, which do not self-propagate but may sum spatially and temporally to trigger an action potential in the postsynaptic neuron.[57] However, neurons actually possess a sophisticated repertoire of responses across spatial, temporal, and voltage scales which enable internal processing and neural computation.[58] These electrical signatures have been documented during patch clamp electrophysiology and include APs, PSPs, backpropagating APs (bAPs), calcium waves, and plateau potentials, which consist of NMDA spike and $Na^+$ spikelet components. Most of these signatures cannot be detected with extracellular microelectrode arrays (MEAs) since they do not enable a transition from extracellular dipole recording[43,59] to intracellular membrane potential recording, as depicted in **Figure 1**. Extracellular electrophysiological recordings using MEAs capture the "local field potential", electrical activity generated from neurons both near and far from the electrode, including putative spikes from single neurons, and neural oscillations generated by the coordinated activity of large populations of neurons.[60] All of these signals can theoretically be obtained using NEAs, including the extracellular LFPs with high signal fidelity.

However, NEAs exhibit less stable electrical coupling[52,31] compared to patch clamping (Figure 1).[31,61,62] This unstable coupling causes signal distortion and attenuation of intracellular signals caused by 1) high electrode impedance, 2) the parasitic capacitance of the amplifier,[35,52,63–65] high electrode capacitive current at the restricted cell/nanoelectrode cleft,[52] and 4) altered junctional membrane resistance.[30,31,66] Underlying this biophysical instability lies variable plasma membranes across cell types,[67,68] electrical field manipulation by nanostructure morphology,[69] and protein redistribution from nanostructure impingement.[18,31,70–81] The dynamic cell/nanoelectrode interface thus makes it difficult to classify events without corresponding ground truth data such as simultaneous patch clamp or pseudo-patch clamp.[1,35] However, achieving paired datasets that match distorted waveforms to their ground truth identity is


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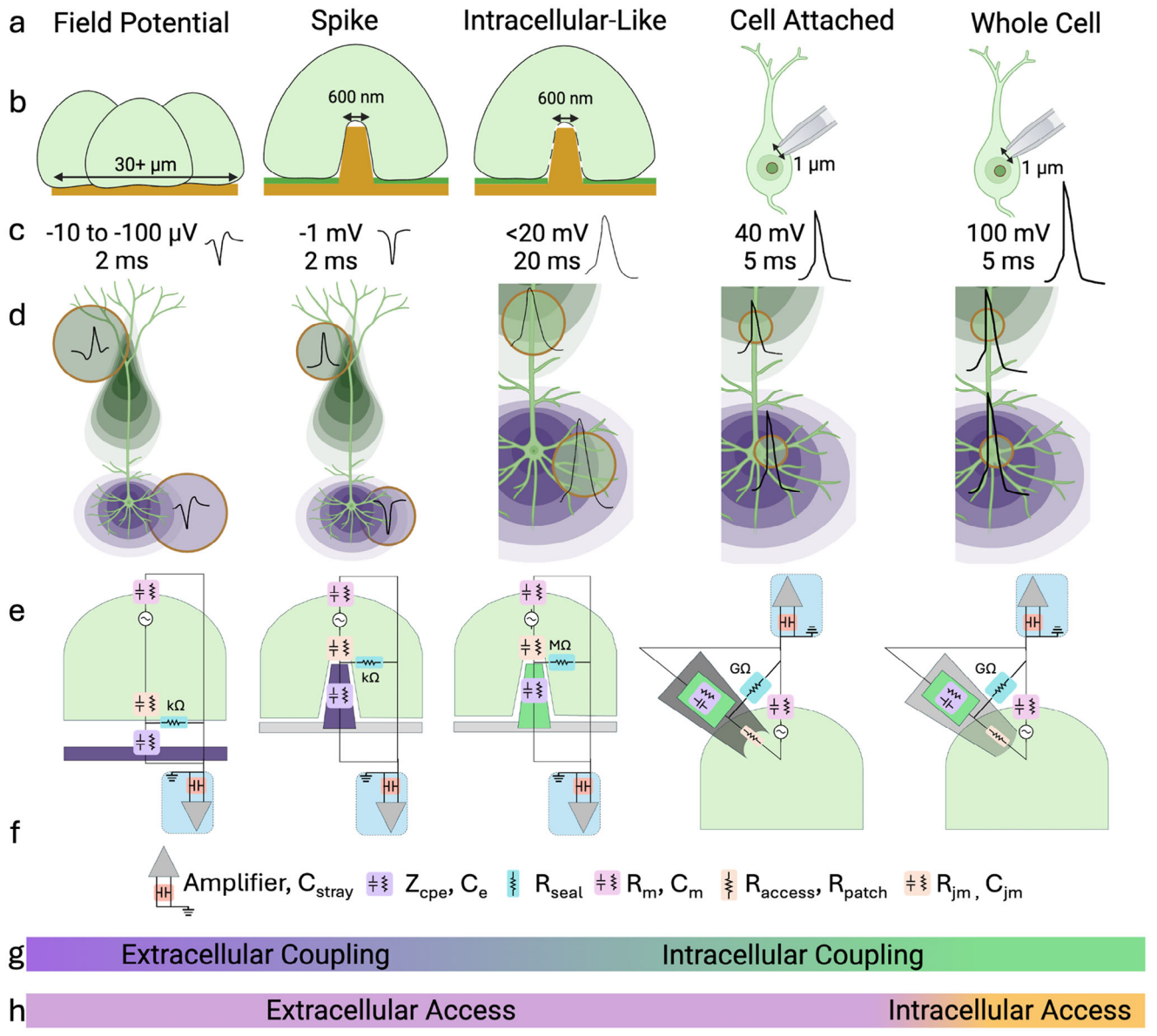


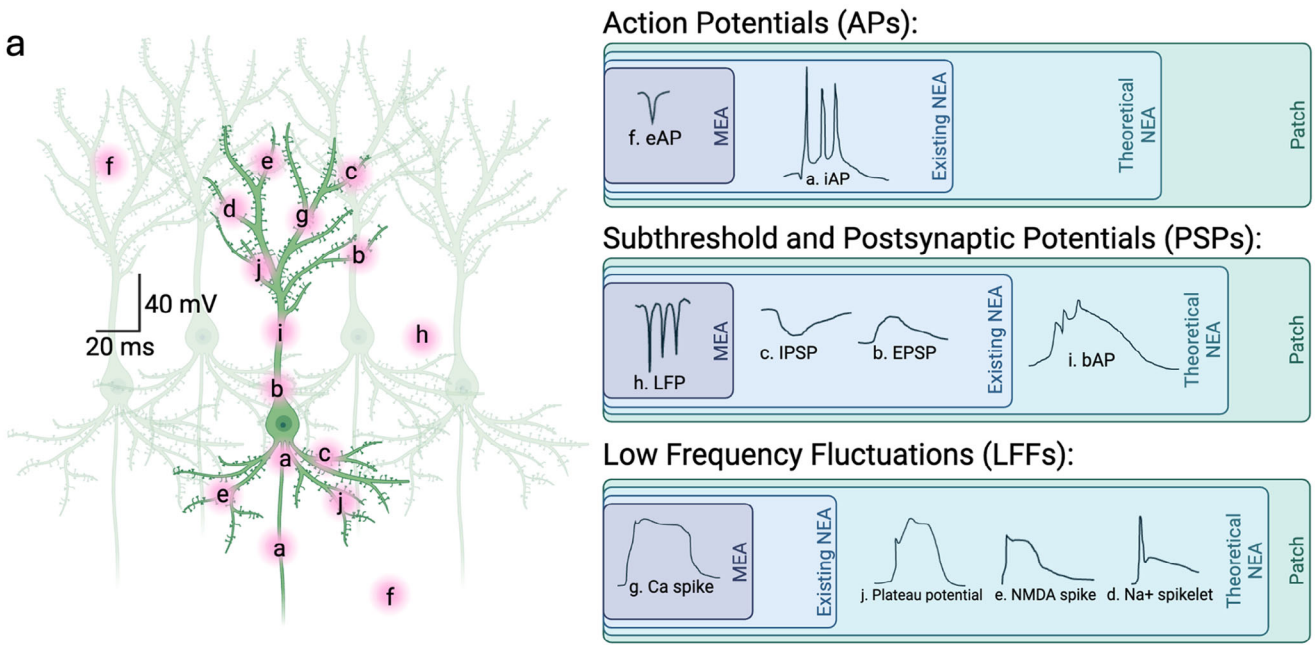

**Figure 1.** NEAs achieve intracellular coupling without intracellular access. The schematic above depicts the continuum of cell-electrode coupling across several electrophysiological recording modalities. The top row in a) indicates the modality displayed, while b) shows the electrode size and proximity to single neurons. c) Depicts the typical amplitude, shape, and duration of action potentials recorded using each of these modalities, with the action potential recorded during whole-cell patch clamp serving as the ground truth.[18,43–49] Note that spike shapes are scaled accurately but not exactly proportional to each other. d) The polarity of the recorded signal changes based on the recording modality. The electric dipole generated by a neuron with an actively propagating AP is shown with the green-shaded (the source) and purple-shaded (the sink) regions. During extracellular coupling, depending on the placement of the electrode, it will record a signal with the opposite polarity as the membrane potential inside the cell at that time (i.e., a depolarized soma will generate a negative extracellular spike). Upon achieving intracellular coupling, the electrode will record a signal with the same polarity as the membrane potential.[43,50] e) Circuit diagrams for each recording modality: field potentials, extracellular spike,[51] intracellular-like NEA,[24,35,44,52–54] cell-attached, and whole-cell patch clamp[55,56] with the legend shown below in (f). g) Scale depicting the transition from extracellular to intracellular coupling. h) scale depicting transition from extracellular to intracellular access. This schematic was created and licensed in Biorender.

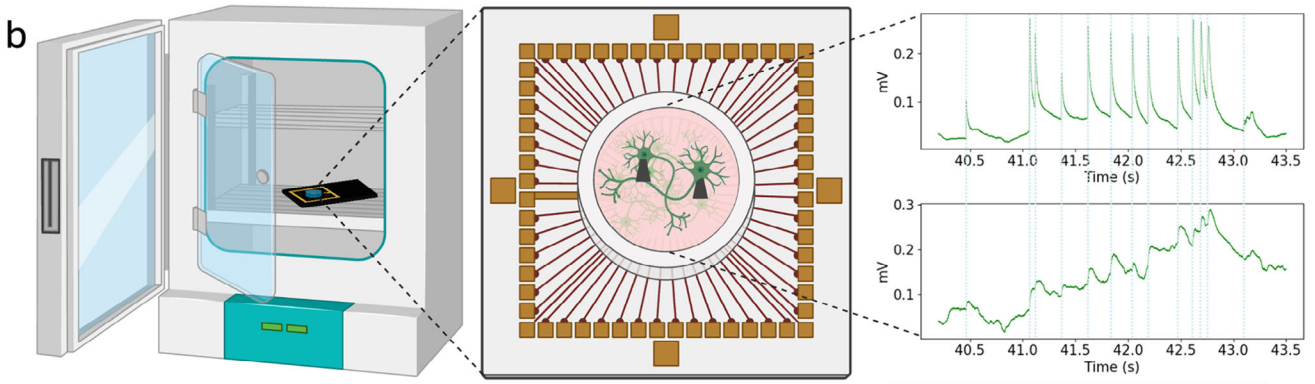
extremely difficult and unscalable.[1,35,52,82] We decided to search for these subthreshold electrical signatures in our own NEA recordings based on the waveform shape, described in detail below.

To assess which of these electrical signatures might be present during NEA recordings, we examined the spatial, temporal, and voltage regimes of these various signatures, and classified them within existing MEA, NEA, or theoretical NEA recording modalities shown in **Figure 2**a. We previously reported a versatile NEA platform compatible with several neuronal cell types and longitudinal incubator recording,[18] shown schematically in Figure 2b. From the example traces in Figure 2b, we observed APs (top channel), PSPs (bottom channel), and also LFFs (both channels) which appeared to correlate with membrane potential states,

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c

| Detects | Processing | Threshold |
|---|---|---|
| APs & PSPs | 5 Hz HP Filter | 5 x std |
| APs only | Raw – 90 ms | 5 x std |
| PSPs only | 90 ms – 370 ms | 4 x std |
| LFFs only | 370 ms | 3 x std |

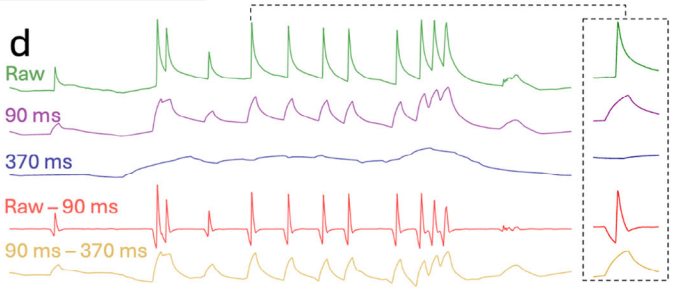



**Figure 2.** Multiscale spike detection. a) Schematic showing diagram of a single neuron with points where various electrical signals can be detected,[58] and the corresponding recording regimes of MEA, NEA, and patch clamp. These electrical signals include spontaneous NMDA spikes,[90–93] which either occur independently or in combination with plateau potentials;[58,94–96] backpropagating action potentials (bAPs),[97,98] calcium spikes,[58,99] and extracellular action potentials.[100] b) Experimental setup for organoid and primary neuron NEA recordings, where NEA device is loaded into headstage inside incubator, and up to 60 channels are recorded simultaneously. The recording traces on the right show an exemplary trace from one channel showing putative action potential firing, with the corresponding postsynaptic potential response in a neighboring channel 320 µm away. c) Table summarizing spike detection methods for supra- and sub-threshold and low frequency events. d) Plot from the top channel in b) showing how multiscale events are detected. The raw trace is run through a moving average filter with either a 90 ms window (to capture subthreshold activity), or a 370 ms window (to capture low frequency activity). Combinations of the differences of these 3 traces are then used to isolate the events within each scale for subsequent event detection. The inset in d) shows a zoomed in trace of a single AP event and the resulting filters below.


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such as the plateau potential which precedes a burst of APs. We then sought to design a spike detector to capture all these types of events.

Typical electrophysiology data analysis pipelines make use of filters to isolate the high frequency component of electrophysiology data which contains the extracellular action potentials. However, there are a few distinctions which make these preprocessing steps unfit for NEA data processing: 1) Common average referencing, which is used as a baseline correction technique across extracellular microelectrodes,[83] cannot be used on nanoelectrodes whose individual transfer functions influence the recorded signal. 2) Filtering, which is used to isolate frequency-tied signal components (such as separating spikes from multineuron LFPs in MEA recordings),[83,84] often employs bandpass filters that do not capture the full spectrum of intracellular signals, or high pass filters which distort certain waveforms due to their near step-like change in voltage (shown in Figure S1b,c, Supporting Information). While Bessel filters or other nearly linear phase infinite impulse response (IIR) filters can dampen this oscillatory distortion, all high pass filters are expected to have some degree of oscillation when filtering a step function and require certain signal conditions to be met.[85–88] Event detection and extraction on high pass filtered waveforms, even with low cutoffs such as between 1–5 Hz, causes waveform distortion into sinusoidal shapes, precluding effective event classification. 3) Spike Extraction, in which extracellular spikes are cut out a few ms before and after the spike crosses the detection threshold, is not compatible with asymmetrical intracellular signals which present a diverse range of durations even within the same signal category. This feature also makes NEA signals incompatible with template matching.[89] 4) Feature extraction and clustering, in which metrics from each spike or the spike waveform itself is used to perform clustering to identify putative single neurons active during recording. This final step of analysis is insufficient to process intracellular data, where single cell status is already known, and subcellular as well as signal categorization is needed instead.

## 2. Results and Discussion

### 2.1. A Dynamic Spike Sorter for Multiplexed, Multiscale Spike Sorting

To extract multiscale events from each channel within our NEA recording datasets, we addressed the above incompatibilities of conventional spike sorters to create a dynamic tool that can analyze multiscale intracellular electrical signals. Our new method, described in **Figure 3**, relies on existing biophysical models[1,52,31] describing the cell/nanoelectrode interface for our bespoke NEAs, and enables the creation of 3D topological maps to better visualize and understand network dynamics within a developing culture. Based on previous research and our own electron microscopic images of neurons interacting with nanopillars (Figure 3a,b),[18,101,102] we knew that the mechanism of intracellular neuronal recording was tight electrical coupling due to engulfment of the pillar by the overlying plasma membrane. Previous research suggests that nanoelectrodes record from multiple locations on the neuronal soma, and high surface area components of primary neurites which wrap around the pillars.[18,35] We performed a CADENCE simulation to describe the expected extra- and intracellular recording modalities using our NEA devices. The circuits and component values used for the simulation (Figure 3c) were created based on similar devices reported previously.[1,52,31] The results of the simulation (Figure 3d) show the expected trend in nanoelectrode technologies: Assuming the non-junctional membrane resistance is much larger than the junctional membrane resistance, signal attenuation and waveform distortion occur to varying degrees depending on the seal resistance. In our model, when $R_{jm}$ = 1GΩ and $R_{jm}$ = 20 GΩ, we modeled a "weak" seal as $R_{seal}$ < 100 MΩ, a moderate seal is $R_{seal}$ = 300 MΩ – 1GΩ, and a strong seal as $R_{seal}$ >1 GΩ. In all cases, the signal is largely attenuated by 40 – 85% (although in our recordings we see ≈ 80% or greater attenuation). However, the signal distortion, which is due to the charging and discharging rate of the junctional membrane capacitance, strongly depends on the seal resistance. We believe this explains why signals within the same event category (i.e., APs) have varying amplitude and duration over several seconds (an example of this is shown in Figure S1a, Supporting Information). This is because the tight electrical coupling opens a resistive "window" into intracellular signals, even when the nanopillar lies outside of the neuron. Not only does this corroborate the findings from other research groups, but it suggested to us that the variable signal fidelity of NEA recordings nevertheless suggests a discrete range of waveform metrics which could be used to categorize multiscale intracellular events. Based on this finding, we proceeded to determine the optimal parameters for isolating and extracting multiscale intracellular signals.

We realized we needed to discard the standard pipeline used by conventional spike sorters for MEA recording (filtering, dimensionality reduction, clustering, and network cross correlation) in favor of an approach which used gentle filters to isolate multiscale signals, cluster these signals separately, and then reorganize them based on their relation with other time-locked events. Figure 3e outlines the final dynamic spike sorting pipeline we used for NEA signal analysis. Based on a technique used during optical electrophysiology,[103] we decided to apply moving average filters that smooth the data to varying degrees, depending on the window size. The advantage of this approach is that waveforms are not distorted, and multiscale components can be effectively isolated for future frequency analysis. Using windows that were roughly the size of the expected PSP (90 ms) and LFF (350 ms) membrane fluctuations, we captured multiscale events by subtracting out quicker events without distorting the true waveform shape or altering the timestamp (due to sinusoidal distortion). Figure 2d above shows an example recording trace with the raw recording in green, a moving average filter with 90 ms window in pink, and a separate moving average filter with 370 ms window in red. We then subtracted the PSP signal from the raw trace to create a flattened signal capturing just APs. We subtracted the LFF baseline from the PSP signal to create a flattened signal capturing just PSPs.

After obtaining these traces containing events across different scales, we experimented with event detection using various thresholds above the standard deviation of the signal. For AP detection, we set the threshold at 5 times the standard deviation, as previously reported.[1] For the PSP and LFF signal detection, thresholds were adjusted to 3$\sigma$ and 4$\sigma$ respectively. Separately,


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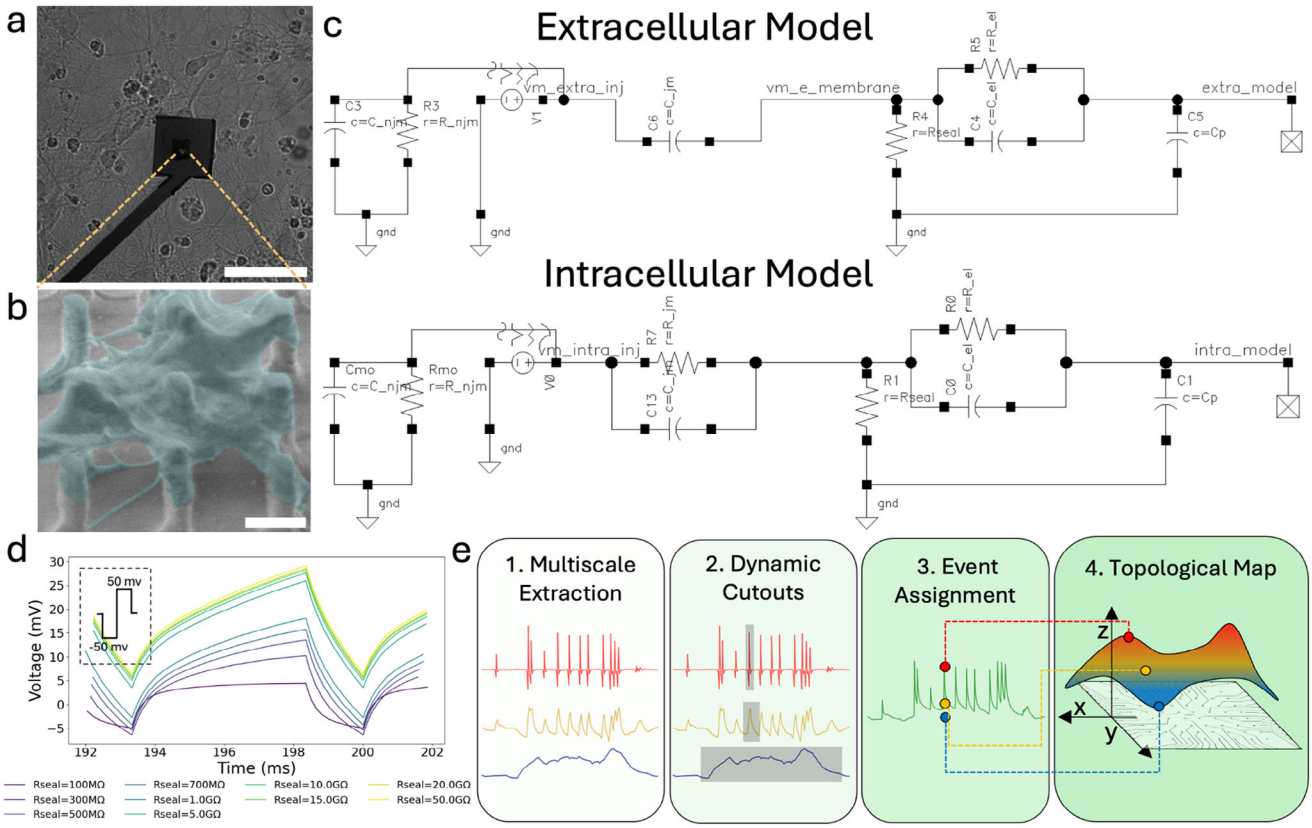



**Figure 3.** Nanopillar electrode interface and simulation. a) Brightfield image showing primary neurons seeded on single nanoelectrode (scale bar = 50 µm). b) SEM image showing neuronal plasma membrane lying on top of nanopillars (scale bar = 2 µm). c) Circuit diagrams used for CADENCE simulation showing extracellular and intracellular NEA recording modalities. d) Simulated voltage trace at the neuron/nanoelectrode interface for different values of seal resistance (inset on left shows injected square biphasic current pulse). e) Schematic showing dynamic spike sorting pipeline for NEA dataset analysis.

we compared the AP detection using the above method to a standard spike detection approach, which is to detect all events above a 5 Hz high pass filter. We saw no clear difference in the number of spikes extracted and so proceeded with the moving average filtered spike detection to remain consistent. The table in Figure 2c summarizes the event detection methods used later in this work. We note that our window sizes resulted in some overlap between events, such as the detection of the slow component of the APs for the first 11 spikes in the pink, 90 ms window trace shown in Figure 2d. The final event in that trace was the only true "PSP" present, which was also detected successfully. Future analysis pipelines may involve re-optimizing the window sizes to more appropriately capture certain types of events, which may also differ across cell types.

### 2.2. Extraction of Multiscale Electrical Signatures

Next, we extracted waveforms from the detected events for subsequent waveform metric and clustering analyses. This is challenging due to 1) the dynamic seal state of single events within the same category, which can cause signal broadening, and 2) the inherent biological variability in temporal duration of the events themselves, both within and across event categories. Because the events detected using the high pass filter may have contained both AP and PSP signals (due to the reported minimum frequency response of PSPs using NEAs being 100 Hz),[1] we also designed a dynamic duration calculator meant to normalize spikes of different durations across the same temporal scale. Specifically, we resampled events which were cut off using recursion on either side of the peak index to all be the same length. This recursion approach involved resetting the left and right indices of the spike cutoff until they reached the baseline noise level of the trace (see Experimental Section). We used the waveform shapes provided by the recursive approach to validate the waveform shapes of the isolated AP and PSP events, shown in Figure S4 (Supporting Information).

For each of the event detection methods, summarized in **Figure** 4a, we created a separate raster plot, shown in Figure 4b. We also plotted the averaged, normalized event waveform shapes for each event extraction method, shown in Figure 4c (note the different time scales on the x-axes), and autocorrelations for each event type, shown in Figure 4d. Raster plots are used in conventional spike sorting analyses pipelines to identify cells which "pace" the culture and may originate spontaneous activity. While this can also be done using just the AP events detected in our NEA recordings, we show the raster plots here to highlight the heavy alignment of shorter events (APs and PSPs) compared to


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a

| Extracts | Duration | Approach |
|---|---|---|
| APs & PSPs | 60 ms or dynamic | 50 ms/70 ms or recursion |
| APs only | 60 ms | 20 ms/40 ms |
| PSPs only | 120 ms | 50 ms/ 70 ms |
| LF only | 1200 ms | 500 ms/700 ms |

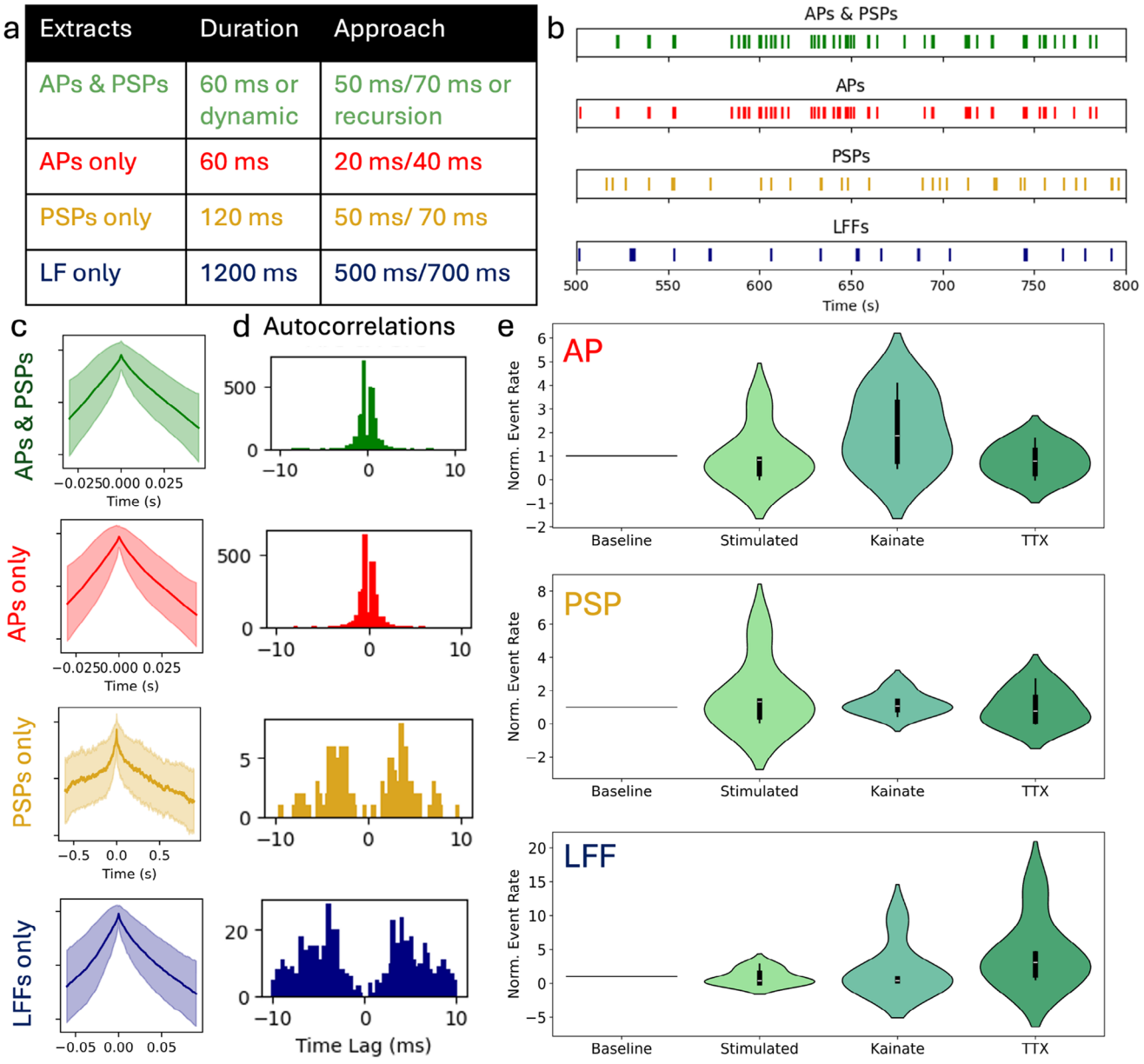



**Figure 4.** Dynamic event extraction. a) Table summarizing multiscale event extraction methods. b) Raster plots from a representative single channel during organoid recording showing rough correlations between the APs or APs and PSPs category, as compared to LFF and PSP event distribution over time. c) Averaged, normalized waveform shapes of extracted events. The bold solid line is the average waveform, and the light-shaded region is the standard deviation across all extracted spikes. This dataset is for all channels recorded during one session with primary hippocampal neurons. d) Representative autocorrelations demonstrate that across all events detected, the delay of the lagged version of these events shows no time lag at zero seconds, suggesting single unit activity. The y-axis units are in Counts, and the x-axis units are in Time Lag (ms) for all plots. e) Violin plots showing event distributions using the full-width-half-max values after various treatments. All plots are plotted along the same y scale within the same row and are normalized along the x-axis.

longer events (LFFs). This is also apparent in the broader autocorrelation histograms in Figure 4c. Qualitatively, it can be seen that i) the detected PSPs seem to precede the other spikes, ii) that there are fewer LFF events, which nevertheless co-occur with other event categories, and iii) that the detection methods including AP events tend to mostly capture APs and not PSPs. More concretely, multiscale raster plots can help pinpoint mechanisms of signal generation (i.e., earlier PSPs suggest that spontaneous activity in this culture originates from subthreshold fluctuations). However, future work should include robust methods to relate rapid events to long events throughout the recording duration, such as studying the spiking rate of shorter APs and PSPs in relation to LFFs, or studying the order of their appearance, amplitude, or duration.

Interestingly, the proportions of event types within single channels for both organoid and primary neuron recordings often

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showed significant differences, shown in Figure S3 (Supporting Information), which may be because of the different neuronal subtypes that can be found within these cultures, and is a future avenue of research. From the event timestamps, we computed the autocorrelations for each of the event detection methods, shown in Figure 4d, which are all bimodal distributions centered around zero. This suggests that, for each type of event – APs, PSPs, and LFFs – a single nanoelectrode is primarily recording from a single neuron. Due to the biological refractory period that follows action potential firing, autocorrelations from single neurons have a characteristic dip to zero around the 0 ms time point.[104,105] Figure 4c shows that the waveform shapes of different kinds of events, particularly that the HP-filtered approach for spike detection mainly captures APs. The autocorrelation histograms themselves show the time scale differences of the multiscale events detected. Again, it is apparent from Figure 4d that the HP-filtered approach mainly captures APs.

After confirming single neuron recording, we investigated whether any waveform metrics could help characterize events during pharmacological interventions and differentiate events which may have been captured by the same extraction method. All following analyses were done on normalized waveforms for each combination of spike detection and extraction method. Previous work suggests that the waveform shape should be largely preserved regardless of attenuation.[21] We computed 1) the area under the curve (AUC), 2) the full-width-half-max (FWHM) –, i.e., the action potential duration at 50% of the total amplitude,[106] and 3) the decay constant ($\tau$) for each of the extracted events. The rapid decay of APs is due to the opening of voltage-gated potassium channels, whereas the slower decay of PSPs is due to a passive (non-propagating) process linked to the plasma membrane time constant.[107] We performed this analysis on data from primary hippocampal neurons under various treatments in succession: First, baseline spontaneous activity – expected to include all events was recorded for several minutes. Next, select nanoelectrodes were given electrical stimulation using a 3.5 V, 200 µs-long pulse (typically used for electroporation, but it had no effect on intracellular signals in this case), which was expected to increase both AP and PSP firing via local membrane depolarization. We previously reported that electroporation with this pulse amplitude and duration does not change the intracellular coupling of the nanoelectrodes with overlying neuronal networks.[18] To achieve membrane permeabilization for overlying neurons with these nanoelectrodes, further studies would be needed to optimize the electroporation protocol.[18,108] Following this, we applied kainate, a glutamate receptor agonist which was expected to mimic excitatory glutamatergic neurotransmission and therefore increase both AP and PSP firing.[109–111] Finally, we applied tetrodotoxin (TTX), a voltage-gated sodium channel blocker which prevents action potential firing in neurons, and therefore was expected to reduce AP firing and indirectly reduce PSP firing.[112] Figure 4e summarizes the normalized event rate responses across single channels for each event, for each treatment. We also investigated whether event waveform metrics, such as AUC, FWHM, or $\tau$ could show the expected trends; however, as shown by their corresponding violin plots in Figure S4 (Supporting Information), there were no obvious trends. Thus, we proceeded to perform dimensionality reduction on the complete waveforms instead of computed waveform features.

Based on the event rate violin plots in Figure 4e, we see many consistencies with the expected results. Compared to the baseline event rate for each channel, electrical stimulation and kainate both increased the number of PSPs present, and TTX reduced the number of both PSPs and APs present during the recording. Interestingly, electrical stimulation decreased the number of APs fired across all channels, possibly due to a "pacing" effect from the stimulation which may have interfered with the native firing rate of the neurons. The LFF events increased throughout the duration of the consecutive treatments, suggesting an increased variability in signal baseline which may be related to noise artifacts emerging during recording for several minutes. The emergence of bimodal waveform distribution in the case of the PSP-only extraction method may be due to the NEA capturing derivatives of "quantized" PSP events which build upon each other before membrane depolarization is achieved.[1,18,35] The multimodal distributions appear in all event types, including in the AUC, FWHM, and $\tau$ measurements in Figure S4 (Supporting Information). In summary, we were able to capture the effects of stimulation and drug treatments using our multiscale event extraction technique and relevant cluster metrics such as the AUC, FWHM, and $\tau$ measurements.

Notably, while the excitatory effects of kainate and electrical stimulation are clear across all of the multiscale events, their relative effects are very different. Particularly, the PSP signal was expected to show a larger increase upon kainate addition, since kainate directly targets kainate receptors responsible for glutamate neurotransmission. This may be due to two reasons. The first is that the PSPs are the signals which lie in the "middle" of the multiscale spikes detected within this work. As such, the time window for the moving average filter and/or the threshold (5 times the standard deviation of the noise) may have caused the PSPs to be underrepresented. In the future, a dynamic threshold based on the underlying LFF could be used instead of a fixed threshold for PSP detection. The second reason for the minimal PSP event increase could be due to a biological effect – kainate should increase both subthreshold and supra- threshold activity, but not necessarily to the same extent depending on the neuron age, cellular subtype type, recording conditions, etc. Future works employing varying concentrations of kainate and/or kainate/TTX cocktails in conjunction with the dynamic spike sorter present in this paper would help researchers quantify, down to the subthreshold level, how different drugs affect neuronal cultures in vitro.

### 2.3. NEA Events are Clustered Based on Symmetry

Following the detection and extraction of multiscale signals from our NEA recordings, we performed dimensionality reduction and clustering. Our primary goal was to determine if the waveforms within event categories contain features reflective of the cell/nanoelectrode seal state, and/or biological states of the recorded neuron. Previous studies have shown that signals can be attenuated or distorted from the seal state at the cell/nanoelectrode interface,[52] but similar waveform changes (AP attenuation and broadening within a burst) are observed during normal bioelectrical activity.[19] To determine if the seal state can influence AP amplitude and width within a burst recorded


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by our nanoelectrodes, we simulated the intracellular and extracellular models in Cadence® software. Our simulation on an example NEA-recorded neuron bursting is summarized in Figure 3. It shows that changing the junctional membrane resistance and/or the seal resistance could cause signal broadening and attenuation, respectively. This varying seal state could therefore result in the variability in the action potential waveform shapes over time (for example, see 160–180 s and after 250 s, Figure S1, Supporting Information). However, even APs that appear well-sealed (i.e., the 3 APs preceding 200 s in Figure S1, Supporting Information) exhibited variability. Notably, some seemed to sit on a "shelf", while others seemed to sit on the "floor" of the baseline activity. Because our recordings were obtained from primary hippocampal neurons, we hypothesized that some derivative of the LFF membrane potential events might capture membrane potential states of the neuron, mirroring the "up" and "down" states or simply membrane potential fluctuations, indicative of excitability, observed in the hippocampus, which are crucial for information processing, sleep and memory formation.[113–117]

Our first step was to reduce the dimensionality of the event waveforms. We used the trustworthiness score[118] to evaluate several dimensionality reduction techniques, including 1) principal component analysis (PCA), a standard approach on neural recording data,[119] 2) non-negative matrix factorization (NMF),[120–123] chosen due to its specificity toward non-negative datasets such as positively-deflecting calcium imaging transients, and 3) t-distributed Stochastic Neighbor Embedding (t-SNE), chosen due to its capacity to reduce datasets with high dimensionality.[124] The trustworthiness score in all cases, across all data subsets analyzed, was above 0.99, so we decided to use PCA to mirror existing spike sorting techniques. Additionally, the percent explained variance did not exceed 70% without 3 principal components (PCs), so all subsequent analysis was performed using 3 PCs instead of 2. Next, we used the K means algorithm for cluster classification and specifically used the Knee/Elbow point detector to determine the optimal number of clusters for each method investigated. This method is widely used in standard unsupervised spike sorters and would allow loose classification of all spikes presented during data processing.[125–127] We also tried using DBSCAN clustering, since it is a density-based clustering method used on datasets where clusters are not spherical.[128] However, we observed that this approach resulted in vastly different clustering classifications across cell types, and required separate optimizations for the initialization parameters. Thus, the clustering approach is a future avenue for investigation but not within the scope of the current work. Figure S5 (Supporting Information) shows representative scatter plots of the first 3 principal components of all primary neuron events extracted using either the cutoff method (top row) or recursion (bottom row). The first column colors each dot by the treatment they received, while the second column colors each dot by the spike time of the event, and the third-row colors each dot by its K-means-assigned cluster. From these plots, it is apparent that the assigned clusters mostly contained waveforms included within each treatment and across the duration of the recording. Interestingly, groups of waveforms within clusters often appeared as "runs" in the PCA plot, and shared very similar spike times, suggesting spikes which occurred in sequence.

We next sought to inspect the waveform shapes of the assigned clusters to determine if they might reflect biological differences in the ongoing state of the neuron. **Figure 5a** shows a representative set of average cluster waveforms from a single channel during a primary hippocampal neuron recording. The clusters are ordered according to increasing AUC value. Upon inspection, we noticed that while there are between 7–9 detected clusters, most of them fell into one of three symmetry designations, either skewed left, skewed right, or symmetrical, depicted schematically and labeled according to Figure 5b. We assigned these symmetry designations the labels of Fedora Left, Fedora Right, or Mexican Hat, respectively, based on existing terms used to describe symmetrical neural constructs.[129,130] Using these assignments on the identified clusters, we examined the cumulative cluster raster plot, shown in Figure 5c, and its corresponding raw recording trace, shown in Figure 5d and colored according to the filters applied in Figure 2. We observed that the symmetry state exhibited by the cluster did indeed reflect the lower frequency membrane potential "state" that the clustered event "sat" on top of. The Fedora left or Fedora right configurations also appeared to correspond to the placement of the event relative to the LFF on which it occurred, that is, on the upward slope or the downward slope of the LFF event. This is depicted schematically using the dashed lines connecting the raster plot occurrences to waveforms within the raw recording trace, and their assigned symmetry state. We suspect that the clustering algorithm, which was performed on the extracted event cutoff waveforms, is largely capturing variability related to the width and skewness of the waveforms. This sensitivity to skewness is why the determined clusters can be related through symmetry. Biologically, skewness likely relates to the local low frequency behavior of the neuron, such as the local membrane fluctuation or position within a burst, at the time of the event. In any case, the "edge effects" which reveal this sensitivity to skewness can potentially help relate APs to underlying membrane potential changes in future studies. Even the edge effects of extracellular spikes have been shown to possess information related to intracellular ionic activity.[82] Using the aforementioned spike detection, extraction, and clustering approach, we believe that NEA recordings can reflect deeper biological interrelationships between high frequency events such as APs, and low frequency events such as LFFs indicative of a single neuron's ongoing functional state within a network.

### 2.4. Creating Topological Maps of Neuronal Network Activity

Finally, we sought to understand how the analysis pipeline outlined in this work could be used to connect subcellular and network activity. Our goal was to investigate how events across the AP, PSP, and LFF event categories would correlate when comparing neurons within the same network. We decided to use NEA recordings of organoids cultured on a device containing paired electrodes, depicted schematically in **Figure 6a–c**. Each of the electrodes contained 3 nanopillars, and the pairs of electrodes were spaced 8.5 μm apart. We determined that the paired electrodes record independently based on the time lag (>2 ms) between each of the events during recording, and that channels not covered by the organoid did not show signal, shown in Figure S6 (Supporting Information). Additionally, this time delay is greater


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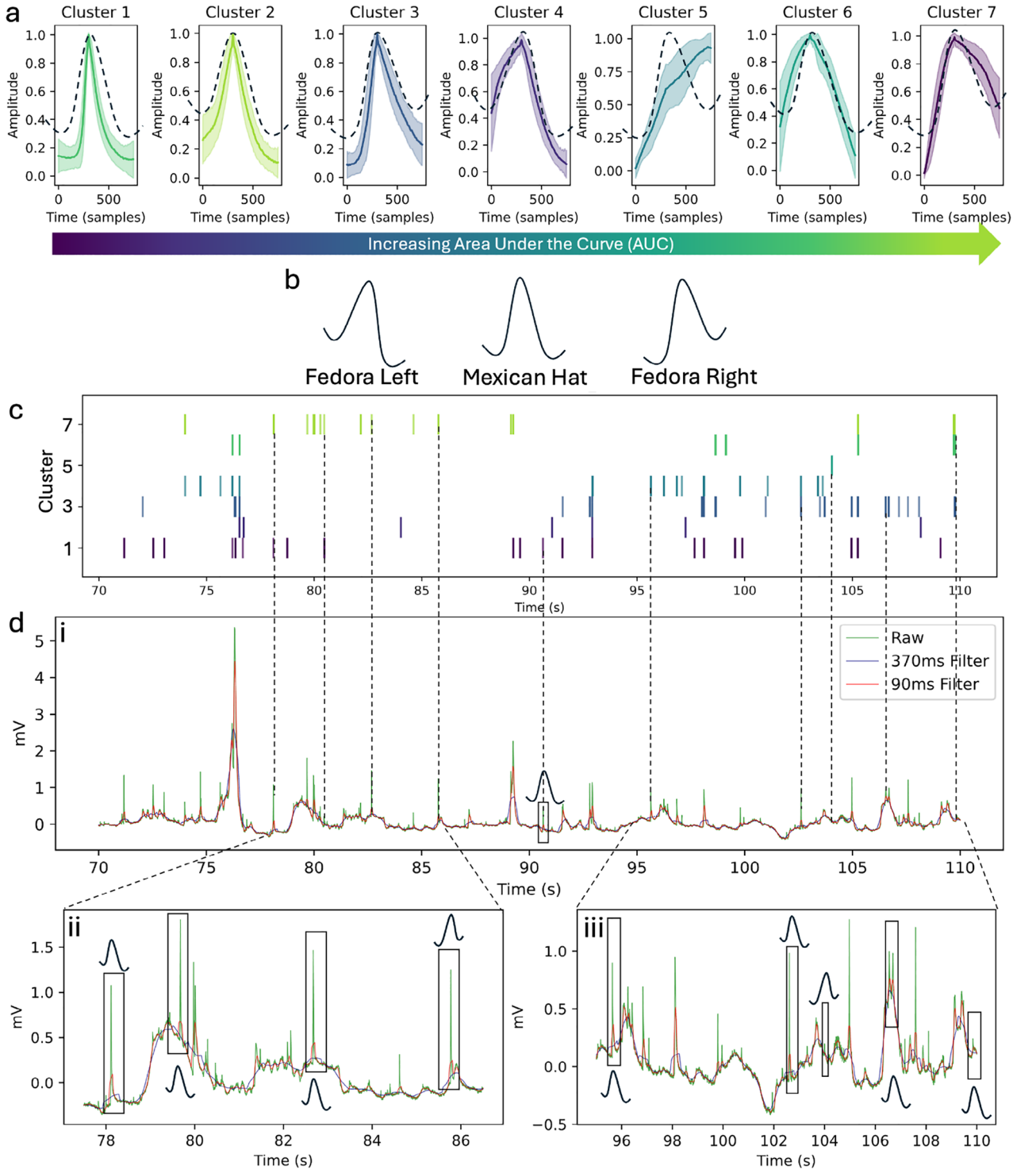



**Figure 5.** Clustering within NEA-detected events. a) K-means clustered waveforms from a single channel during primary hippocampal neuron. Waveforms are ordered by increasing area under the curve (AUC) value. The dotted black lines superimposed onto each waveform indicate their classification into 3 categories. b) Representative traces of the 3 identified categories to classify types of clusters: Mexican Hat, Fedora Left, and Fedora right. c) Raster plot corresponding to the baseline recording for the trace used to create the waveform clusters in a). The clusters in c) are now colored according to increasing AUC value. d) i) Raw baseline recording trace with 90 ms and 370 ms moving average filter windows superimposed on top. ii) and iii) show the same trace zoomed in to clearly show event occurrence on top of an uneven baseline. The black boxes outline events that correspond to their position in the raster plot in c) using the black dashed lines. The putative categories, taken from b), for each of these events are shown with the small hat-like icon.


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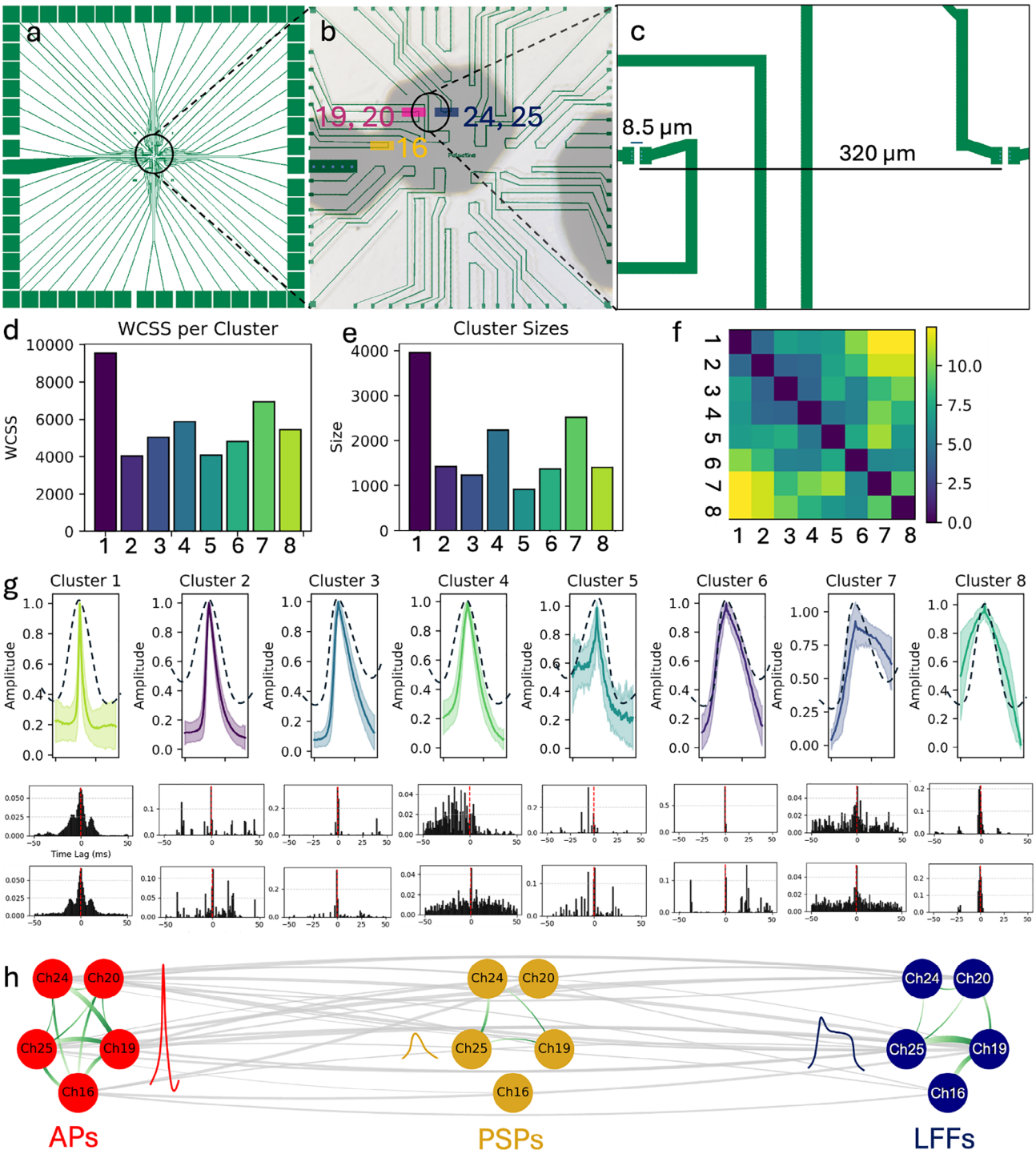



**Figure 6.** Sub-AP-event clusters influence network behavior. a–c) Schematic showing NEA device with paired electrodes, designed to capture pairs of neighboring neurons. The labels in b) indicate a close channel (24) and a neighboring channel (20) compared to the target channel 25. d–f) Cluster metrics after K-means clustering of the entire organoid recording from this device. d) and e) show the Within Cluster Sum of Squares, which is the sum of the variance between the observations in each cluster, e) shows the cluster size, and f) shows the inter-cluster distance. g) shows the K-means clustered waveforms from this recording along with their superimposed symmetry categories, ordered in terms of increasing AUC value. Below each cluster plot, two cross correlograms comparing channel 20 to 25 (neighboring – top), or channel 24 to 25 (paired – bottom) are shown. h) Nodal network across 5 channels showing within-event (green gradient curves) and across-event (grey curves) peak time lags falling within 20 ms of each event timestamp.


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than the expected synaptic time delay between neurons, suggesting that paired electrodes are recording from adjacent neurons and not from the same neuron. We detected all the events across all 5 channels showing activity and performed PCA and K-means clustering as previously described above.

Upon obtaining the clusters, we investigated the intra- and inter-cluster metrics, summarized in Figure 6d–f, for the clusters identified using K-means in Figure 6g, to determine whether certain clusters were similar and therefore likely to have a similar biological origin. The first and smallest (based on AUC value) cluster detected showed the highest within-cluster sum of squares, indicating high cluster compactness, and the highest cluster size. It also showed the greatest likeness to the Mexican hat designation discussed above. We saw a similar trend for the first cluster across all channels and cell types evaluated using this approach. When matched to the raw trace, the events from this first cluster "sat" on the floor of the recorded baseline. This and the smallest cluster's high frequency suggests that it may be a multiple of the minimum number of synaptic vesicle quanta for PSPs that the NEAs are sensitive to[131] instead of APs, since APs are unlikely to occur during hyperpolarized states. We next searched for trends across the other clusters which tended to be asymmetrical.

Based on the inter-cluster distance matrix shown in Figure 6f, it was clear that clusters with similar AUC values tended to collect closer together when considering the first 3 principal components. We reasoned that this trend was likely due to clusters with similar AUC values resulting from similar seal states during recording. However, this did not necessarily hold true for all the clusters; clusters with the same skewness also tended to be closer together. These two observations suggest a progression of spikes throughout different seal states, which retain the same symmetry but get distorted (broadened) as the cell/nanoelectrode interface changes. We saw further evidence of these observations in the cross-correlograms shown below each cluster waveform in Figure 6g.

Specifically, we observed that correlations between channels within the same cluster showed similar connectivities. The two Mexican Hat-classified clusters, Clusters 1 and 8, show few-millisecond temporal correlations when comparing neurons from an adjacent (channel 24) and far (channel 20) channel to the channel of interest (channel 25). Based on the cross correlograms, channels 20, 24, and 25 appear to be synaptically connected, with channel 24 firing first, followed by channel 25 and then channel 20. The cross correlograms for the other clusters show a similar trend. We also investigated if the cross correlograms across events (i.e., APs, PSPs, and LFFs) were also correlated. These plots are shown in Figure S7 (Supporting Information) along with the corresponding raster plots for each cluster within each event. They suggest that PSP and AP events converge with decreasing distance to the channel of interest, but LFFs remain independently distributed and may reflect the neuron's own biological state rather than its role within the network.

Finally, we wanted to demonstrate that intracellular-like recordings, such as those obtained using NEAs, can be used to create topological maps of network activity across voltage and temporal scales. This proof-of-concept map is shown with this dataset in Figure 6h. The nodes representing each channel are shown in 3 different event layers: APs, PSPs, and LFFs. The edges connecting the nodes within each layer are shaded green and indicate functional connections (firing within 20 ms) between channels. The within-event edge thickness indicates the positive time lag between channels. The node connected to the lighter part of the edge fired before the node connected to the darker part of the edge. Separately, the edges connecting the nodes across each layer are shaded light grey, and indicate that even across tens of milliseconds, events from different channels/neurons show high correlations. This topological map demonstrates that the stratification of event activity can be used to discern ultrastructural trends during neuronal network function using highly sensitive intracellular biosensors.

## 3. Conclusion

In this work, we outline an analytical approach to detect, extract, and classify multiscale electrical events from NEA recordings that overcome technical limitations of conventional pipelines primarily designed for spike sorting of extracellular action potential data. We show that we can detect several different classes of electrical events including high frequency AP and PSP events, as well as low frequency LFFs which reflect membrane potential state, which is made possible by the intracellular coupling achieved by extracellular nanoelectrodes underlying neuronal networks. This finding is validated in both cultured primary neurons as well as neural organoids, and their separate pharmacological and paired channel data suggest that isolation of multiscale signals provides a rich landscape to study subcellular and network dynamics.

Although these events are partially distorted and attenuated due to current losses at the dynamic cell/nanoelectrode interface, these event types can be reliably detected and extracted using this dynamic spike sorter. Notably, replacing high pass filtering with gentle moving average filters preserves the biological features within waveform shapes, as exhibited by symmetry classifications of AP event clustered waveforms. The application of adaptive thresholds for each multiscale event category ensures precise event detection and extraction. The positioning of AP and PSP events can be captured by the edges of extracted events, and reveals how waveform shape changes within bursts of heightened activity. Across event types, both AP and PSP events appear to be affected by the neuron's placement within the network, while LFF events appear to occur independently within single neurons.

The main limitations of our approach include insufficient analysis to relate rapid events (APs and PSPs) to slow oscillations (LFFs) and extract meaningful correlations between the subcellular function and the network activity. Drawing upon multi-modal analysis approaches where the timescales of different modalities vary could improve future dynamic spike sorters.[132] Also, due to the need to manually identify the window size for gentle filters across cell types, improved spike sorters might include frequency analyses to automatically identify the correct window size for each event type. Future investigations for this pipeline should include systematic studies of event proportions to delineate either neuronal subtype or neuronal function within a network. Instead of clustering based on waveform signal, robust metrics based on frequency analysis, relative positioning to other multiscale events, and subcellular location could be used to better delineate events. This could lead to real time event classification and correlation during live recording sessions, and investigations of subcellular – network links using diseased tissue.[133,134]


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Investigating the relationships between clusters associated with different biological mechanisms would also be of interest. Understanding how postsynaptic potentials correlate with action potentials will be of great value in understanding neural computation, and how this changes over time in the case of synaptic plasticity. Furthermore, alternative approaches to investigate how action potentials integrate with the ongoing state of the neuron, reflected by the low frequency fluctuations, will be valuable to determine whether these low frequency fluctuations are truly reflective of the excitability state of the neuron. We believe this work holds promise for development of future iterations of dynamic spike sorters that allow neuroscientists to leverage the potential of nanoelectrode arrays and create maps of multiscale activity across a wide range of neuronal networks.

## 4. Experimental Section

Nanoelectrode array (NEA) fabrication, cell culture, and electrophysiology were performed as previously described.[18] Briefly, nanoelectrode arrays were fabricated in the Nano3 cleanroom facility at the University of California San Diego (UCSD). First, nanopillars were built on 4″ fused silica wafers using two step dry and wet etching. Next, a microelectrode array pattern was sputtered on top with Ti-10 nm/Pt-40 nm and passivated using SiOx/SiNx alternating layers. After dicing, plastic or glass wells were glued onto the NEA chips and the devices were sterilized using 70% ethanol and UV light. The appropriate protein coating was deposited onto the devices as previously described,[18] depending on the desired cell type for recording. In this work, only recordings from hiPSC-derived brain organoids from the lab of Prof. Alysson Muotri, or primary hippocampal neurons from mice from the lab of Prof. Sreekanth Chalasani were used. All recordings were performed by clicking the NEA device into a single, sterilized MultiChannel Systems headstage with the MultiChannel Systems Ag/AgCl pellet external reference electrode, and placing the headstage inside the incubator.

All recordings were collected using the MultiChannel Systems equipment and saved as. hdf5 files, as previously reported.[18] They were loaded and concatenated across consecutive recording sessions (in case the software crashed) using Python supported by Google Colaboratory in a high RAM environment. Raw recordings were performed at 25 kHz sampling rate, but were downsampled to 10 kHz and passed through a moving average filter with 1 ms window before any processing. Spikes were detected by applying adaptive thresholds (3, 4, and 5 times the standard deviation of the RMS noise for LFFs, PSPs, and APs, respectively) to each of the isolated signals. Then, the indices for the detected spikes for each multiscale event category were used to cut out events from the raw data with only a 1 ms moving average filter applied (hereinafter called the Raw+1MAF data). Or, recursion was used to extract the events from this raw data with the 1 ms moving average filter. During the recursive approach, the total amplitude of an initial spike cutoff was computed. Then, a certain percentage (either 50% for the left side of the spike peak or 33% for the right side of the spike peak) of the amplitude was calculated. The left or right index was repositioned at this new temporary amplitude, assuming the drop in amplitude did not drop below the RMS noise level of the signal. This process was repeated on either side of the spike until the index dropped below the noise.

For dimensionality reduction and clustering, several methods were used using existing Python libraries. These two steps were applied to the Raw+1MAF event cutouts for each multiscale event category. Based on the trustworthiness score, which was the same for all dimensionality reduction methods used, PCA was used to maintain consistency with other spike sorters for eventual comparison. Based on optimization of parameters related to the number of clusters, cluster strength and closeness, Kmeans was selected as the best approach for clustering. Clustering and event rate analyses were performed on organoid and primary neuron data separately, since the data represented different experimental paradigms (i.e., paired channel recordings versus pharmacological recordings, respectively). Metrics to characterize waveforms such as area under the curve, decay constant, amplitude, and duration (such as the Full Width Half Max value) were calculated in the Python code itself for each event from the Raw+1MAF cutout. All plots were created in Python. ChatGPT was used to aid algorithm development during this project. The code and example data are available at https://github.com/Bionelab.

*Ethics Approval Statement*: All experiments and protocols outlined above were approved by the Institutional Animal Care and Use Committee of the Salk Institute for Biological Studies and the Institutional Review Board at UC San Diego.

## Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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## Conflict of Interest

The authors declare no conflict of interest.

## Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

## Keywords

multiscale, nanoelectrode arrays, neuronal networks, spike sorting

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